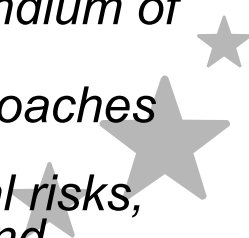


Good practices for advancing climate and nature-related risk management

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ECB Banking Supervision has updated its compendium of good practices for climate and nature-related risk management and stress testing. It leverages approaches that are already applied by more than 60 different institutions to close gaps in areas such as physical risks, prudential transition planning, scenario analysis and nature-related risks. To ensure that the compendium is also useful for smaller and less exposed banks, we show good practices with different levels of sophistication.



Banks have improved their practices for managing climate and nature-related (C&N) risks in recent years. All banks under our supervision now have the foundational architecture in place to identify, quantify and manage the risks from the ongoing climate and nature crises. These improvements are not only vital for ensuring resilience but also enable banks to seize the business opportunities presented by the green transition, benefiting their competitiveness.

Notwithstanding this clear progress, more still needs to be done to ensure banks apply sound practices across all material portfolios, exposures and risk categories.^[1] The methods used, especially for measuring physical and nature-related risks, are still in their infancy with risks very likely being underestimated. Some bank-specific weaknesses also still need to be addressed.

What's more, the evolving risk landscape with high uncertainty makes C&N risk management even more challenging. We are heading towards a disorderly transition scenario with higher uncertainty. It is therefore crucial to be resilient and prepared for a range of possible scenarios – including higher and faster-moving transition and physical risks.

Why we update good practices

ECB Banking Supervision is updating its compendium of good practices for [C&N risk management](#) and [stress testing](#) to support banks with know-how to close gaps in their risk management frameworks and address vulnerabilities in the evolving risk environment. Building on our unique vantage point as pan-European supervisor – where we can look closely inside the cockpits of different banks – good practices are drawn from approaches already applied by more than 60 different institutions, representing more than half of the institutions directly supervised by the ECB.

The good practices compendium builds on extensive supervisory dialogue with banks, in which banks repeatedly asked for insights regarding good management of climate and nature-related risks. This compendium therefore aims to serve different functions within the banks, by describing good practices in sufficient detail and placing particular focus on areas where approaches are typically less advanced, such as physical or nature-related risks. The updated good practices also support in a proportionate way smaller and less materially exposed banks by pointing to less sophisticated practices as well as existing and publicly available tools that other banks have benefited from.

Considering that we are currently reviewing our supervisory guides^[2] let me highlight again that good practices have no legally binding effect and do not describe or establish new legal or regulatory requirements. As such, good practices are not a prerequisite for banks' compliance with the applicable legal framework. Instead, they are intended to provide a range of examples that banks' may take inspiration from when building robust C&N risk management capabilities in light of the [European Banking Authority \(EBA\) Guidelines on the management of Environmental, Social and Governance \(ESG\) risks](#) [environmental scenario analysis](#) as well as the [ECB Guide on climate and nature-related risks](#).^[3] The examples of measures and practices described in this report are illustrative only, and are not intended to be exhaustive. This means that a bank may be fully compliant with the applicable legal framework without implementing any of the specific examples set out in this report, provided that it has implemented other measures and practices that are more appropriate to its particular risk profile, business model and circumstances.

In the updated compendium we have placed particular emphasis on areas where banks typically struggle with, for example quantifying physical risk.^[4] Observed good practices go far beyond physical risks. The examples of prudential transition planning, scenario analysis and nature-related risks – all of which pose distinct challenges – show that some banks already apply a wealth of good practices even in more demanding areas.

Using transition planning to manage risk under uncertainty

Managing risk in a highly uncertain environment is a difficult task. We simply cannot predict the exact level of transition and physical risks we will face. These are influenced by policy decisions, technological developments and global geopolitical dynamics. However, what we do know is that we will be affected by *some* combination of these risks. Faced with uncertainty and geopolitical tensions, it is all the more important that banks are capable of charting a steady course through these changes and remaining resilient in different scenarios.

Prudential transition planning supports banks in understanding how different plausible transition pathways can affect their risk profiles, ensuring they remain resilient over the medium to long term. The regulation on prudential transition planning is not prescriptive as to what specific decarbonisation pathway banks are supposed to be on. Instead, it requires banks to think consciously about their strategy in light of these decarbonisation pathways, set their own objectives in the long term and have an efficient risk management framework in place.

Prudential transition planning practices are still uneven across banks and still developing in a number of areas. We will therefore continue to address this topic in our supervisory dialogue in the months ahead.

I would like to highlight below three helpful examples from our compendium of good practices.

Transition finance

Some banks use their extensive technological knowledge of hard-to-abate sectors, such as cement, steel and aviation, to design transition finance strategies and products that help their clients make the change from high-emission to low carbon technologies. Rather than abandoning the relationship with clients in these sectors, banks can capitalise on the transition by providing greater volumes of transition finance and diversifying their income streams by offering advisory services. This creates a win-win situation whereby clients are supported on their decarbonisation pathways and banks can strengthen their competitiveness.

Client engagement to manage physical risks

Banks have started to incorporate physical risks in their prudential transition plans. We see good practices, in particular, where banks actively engage with their clients. While some banks may use blanket approaches, such as higher pricing for entire regions and sectors, or retreating from these altogether, we see banks increasingly adopt more sophisticated approaches by actively engaging with corporate clients in high physical risk sectors. Banks focus for instance on physical risk impacts on clients' collateral. This client-specific way of addressing physical risks in bank portfolios may well be more risk-sensitive and effective than blanket approaches.

Strategic pricing and profitability steering

Having identified specific transition technologies, such as renewables, as a strategic growth area, some banks realise that the pace of technological and business model change means that some projects do not straight away meet their profitability targets. They are therefore prepared to accept lower margins in the short term, while investing in internal expertise, data infrastructure and client relationships over a longer-term horizon. This allows banks to develop a strong position in a growing market, thereby supporting long-term profitability. Banks are also developing targeted transition finance products and use pricing incentives. In addition, banks are becoming more involved in providing finance to startups, for example through venture debt instruments.

Scenario analysis and stress testing

Scenario analysis and stress testing are indispensable tools for managing risk in times of heightened uncertainty. Despite the progress banks have made, there is still more work to do to improve estimation methodologies and broaden the risk management toolkit.^[5]

The good practices outlined in our compendium can help banks overcome practical challenges and further strengthen their climate and nature-related scenario analysis and stress testing – also in view of the EBA Guidelines on environmental scenario analysis, which take effect in early 2027.^[6]

Let me highlight two promising examples of good practices in scenario analysis and stress testing.

Transition risk modelling at the counterparty level

Instead of relying solely on broad sectoral assumptions, some banks are assessing transition risks at the individual client level. They first estimate how scenario-specific developments – such as higher carbon prices or green investment needs – affect a company's financial ratios. Next, they feed these indicators into existing internal rating systems to derive stressed probabilities of default. This means that they can make a more granular distinction between individual client risks in sectors where vulnerabilities depend heavily on each company's energy mix, technology and transition strategy.

Assessing acute physical risks with more granularity

When assessing acute physical risks, some banks are moving away from broad regional classifications^[7] and towards mapping exposures to the exact geographical locations of individual buildings or assets. One key advance is the use of internal vulnerability indicators combining external hazard maps with client data. These indicators include a customised drought risk index that merges open-source hazard data with client-specific water-intensity metrics. Some frameworks even go a step further, modelling how extreme weather events disrupt corporate operations. By simulating the duration of business interruptions caused by events such as floods or wildfires, banks can derive stressed probabilities of default along with loan-to-value and loss-given-default metrics directly from the resulting declines in revenue and physical damage to collateral.

Nature-related risks

Nature-related risks are an area where approaches are more in their infancy.^[8] While most banks have performed materiality assessments, around two-thirds of them do not yet systematically link these to risk management actions. Banks have extended the set of risk management tools beyond exclusion criteria to client engagement, and in some cases quantitative approaches. However, only a few banks have defined nature-related key risk indicators. Where such indicators exist, they are often used solely for monitoring, with no limits or thresholds to trigger management action.

To help banks make progress in this area, around one-third of the new good practices in our compendium are focused on nature-related risks. The compendium does not simply provide a snapshot of what the frontrunners are doing but instead illustrates, step by step, different ways in which banks can monitor, manage and mitigate the nature-related credit risks to which they are most exposed.^[9]

Using a broad range of public tools and datasets

The compendium covers a wealth of publicly available tools and datasets. These range from general nature indicators, for example assessing companies' dependence on ecosystem services or their impact on biodiversity, to more targeted indicators, such as their geographical distance from protected areas, pollution levels and water consumption. Tools of this kind are helping banks to move from qualitative assessments to quantitative approaches. For example, some institutions are quantifying nature-related financial impacts using scenario analysis for clients operating in the most sensitive sectors and geographies. This enables them to test the resilience of these clients to nature-related risks such as water scarcity, the introduction of environmental permit systems and pollution tax.

Client engagement and due diligence

Proactive client engagement is one of the tools most commonly used to manage nature-related risk exposure. The compendium describes assessment criteria and engagement policies, risk driver by risk driver to support sound due diligence and client engagement. We show how banks collect relevant data, engage on specific risk exposures and offer nature-related financial products or advisory services. By working with clients on their transition – rather than simply withdrawing from the relationship – banks can ensure clients’ practices are aligned with their own risk policies and strategic approaches. This ultimately reduces banks’ exposure to nature-related risks.

Integrating nature-related risks into banks’ internal capital adequacy assessment processes

Some banks are now integrating nature-related risks into the risk inventory and risk management elements of their internal capital adequacy assessment processes (ICAAPs), conducting nature-related scenario analyses and in some cases defining nature-related capital buffers to cover potential losses.

Next steps

Euro area banks have made significant strides in building up their resilience to C&N risks, but the journey is far from complete. The good practices compendium is an effective toolkit for banks to tackle gaps, navigate a challenging and evolving risk environment – and in turn – capitalise on the opportunities offered by the transition. As part of our ongoing supervisory dialogue, we will continue to work closely with banks to make sure they are resilient to C&N risks, particularly at a time of heightened uncertainty, in line with our supervisory priorities for 2026-28.^[10] This is all the more important given the mounting evidence that the forward-looking nature of these risks – for example non-linear dynamics and compounding events when measuring physical risk – are still not fully understood. As a result, such risks may be underestimated within the financial system.^[11] The growing insurance protection gap,^[12] together with the strain on public finances that is weakening governments’ role as “insurer of last resort” for climate and nature-related losses, may lead to C&N risks having an increasing impact on banks’ balance sheets.

Against this backdrop, we will place even greater emphasis on the topics of risk underestimation, physical risks and C&N risks under heightened uncertainty to ensure resilience. Because only resilient banks can play their vitally important role in the economy, financing investments in the green, digital and defence transitions, so that as Europeans we remain masters of our own destiny.

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1.

For example, some banks may focus only on transition risks but not on physical risks – and vice versa – in certain portfolios. Some prudential transmission channels, such as operational or market risk, receive far less attention compared with credit risk. Furthermore, not all banks have included all material risk drivers, relevant portfolios or transmission channels in their C&N risk management. For more details on the lack of comprehensive coverage see Elderson, F. (2025), “[Banks have made good progress in managing climate and nature risks – and must continue](#)”, *The Supervision Blog*, ECB, 11 July; Elderson, F. (2025), “[From charting the course to staying the course: the path ahead for climate and nature risk supervision](#)”, keynote speech at the ECB industry dialogue on “Climate and nature risk management: taking stock and looking ahead”, Frankfurt am Main, 1 October.

2.

As part of its “Next-level supervision” simplification project, ECB Banking Supervision is currently reviewing its supervisory guides. The review aims to ensure that supervisory guides remain fit for purpose, transparent and consistent with the legal framework. For more information, see ECB (2025), “[Streamlining supervision, safeguarding resilience: the ECB’s agenda for more effective, efficient and risk-based European banking supervision](#)”, December.

3.

For more details on supervisory guides, in particular the difference between binding obligations derived from prudential regulation, non-binding supervisory expectations and good practices, see Elderson, F. (2025), “[Making supervision simpler: the role of supervisory guides](#)”, Florence, 27 October.

4.

For example, the compendium’s section on physical risks details public – often EU-funded – datasets to quantify physical risks and assess capital needs (for example the Copernicus Climate Change Service and C3S Climate Data Store, and the DRMKC Risk Data Hub from the Joint Research Centre) as well as structured processes for acquiring data on physical risks from third-party vendors.

5.

For instance, some banks do not include all material risk drivers, relevant portfolios or transmission channels in their scenario analysis and stress testing. This suggests that risks may still be underestimated.

6.

EBA (2025), [*Final report on Guidelines on environmental scenario analysis*](#), November.

7.

See “[NUTS – Nomenclature of territorial units for statistics](#)” on Eurostat’s website. NUTS 3 refers specifically to small regions.

8.

For more information on the particular challenges posed by nature-related risks, see Elderson, F. (2026), “[Nature in decline, economy on the line: the importance of international cooperation for managing nature-related risks](#)”, opening remarks at the at the NGFS Annual Plenary Event panel discussion on “Incorporating nature into supervisory practices”, Frankfurt am Main, 9 March; and Elderson, F. (2025), “[Nature’s bell tolls for thee, economy!](#)”, keynote speech at the Naturalis Biodiversity Center, Leiden, 22 May.

9.

For example, banks are increasingly adopting a layered approach to nature-related risks. Alongside a general framework for monitoring, managing and mitigating nature-related risks – often together with climate-related risks – they have targeted policies that focus on specific risk drivers, geographies and sectors. This has allowed banks to address the most pressing nature-related exposures in greater detail.

10.

ECB (2025), “[Supervisory priorities 2026-28](#)”, 18 November.

11.

See Network for Greening the Financial System (2024), “[Acute physical impacts from climate change and monetary policy](#)”, Technical document, August. The document discusses macroeconomic effects that arise from the compounding of multiple severe weather events and their interaction with the chronic effects of climate change. See also Institute and Faculty of Actuaries and the University of Exeter (2025), “[Planetary Solvency – finding our balance with nature. Global risk management for human prosperity](#)”, January; and University of Exeter (2026), Abrams, J.F., Hu, S. and Bampton, B.D. (2026), “[Recalibrating Climate Risk](#)”, February.

12.

As risks continue to grow, insurance coverage for natural catastrophes remains insufficient. For example, only €4.5 billion of the €11 billion in losses from the Valencia floods in 2024 were insured, and only €13 billion of the €51 billion in losses from the 2021 floods in the Ahr valley were covered. See EIOPA and ESM (2026), “[Sharing the risk: a European approach to natural catastrophe risk management](#)”, April; and Financial Times (2026), “EU urged to create €10bn-€65bn fund for natural and climate disasters”, 9 April.

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